

sellsell marketplace updates - 13042026

UPDATE: 13/04/2026:05:57PM

SHAZ EJ



SellSell / PickPic — Project Overview & Progress Report

1. Vision of the Project

SellSell is an **AI-powered marketplace platform** designed to transform buying and selling into a **human-like conversational experience**.

Instead of traditional keyword-based marketplaces (like Haraj, Q84sale), SellSell enables users to:

- Search using natural language (“cheap family car under 3000”)
 - Interact through chat like a human assistant
 - Automatically generate listings using AI
 - Discover products intelligently based on intent
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2. System Architecture (High-Level)

The platform is built as a **full-stack AI marketplace system**:

Frontend

- Next.js (App Router)
- Chat-based UI + listing UI
- Admin dashboard

Backend

- Next.js API routes
- Prisma ORM

Database



- PostgreSQL (structured data)
- Redis (sessions, queues)

AI & Search

- Ollama (local AI inference)
- Qdrant (vector database for semantic search)
- Hybrid search (vector + keyword BM25)

Storage

- S3-compatible storage for images

3. Core Systems Built

3.1 Smart Data & Enrichment Layer

Problem

Raw marketplace data is unstructured and hard to search.

Solution

Built a **Smart Data Enrichment System**:

Each product now includes:

- smartCategory
- priceBand (cheap / mid / premium)
- intentTags (family, luxury, budget, etc.)
- completenessScore



Impact

- Enables **intent-based filtering**
 - Powers better ranking
 - Improves search accuracy dramatically
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3.2 AI-Powered Search System

Features

- Natural language understanding
- Intent parsing:
 - “cheap” → priceBand = cheap
 - “family car” → intentTags = family
- Hybrid search:
 - Semantic (vector)
 - Keyword (BM25)

Ranking Engine

Custom ranking logic combining:

- Semantic similarity
- Price relevance
- Intent matching
- CompletenessScore
- Freshness

Result

Search evolved from:

“keyword matching” ❌



to
“human-like understanding” ✓

3.3 Conversational AI (Chat System)

Features

- Multi-turn conversation memory
- Context merging (e.g., “cheap car” → “SUV” → “under 3000”)
- Ambiguity detection:
 - “apartment” → rent or buy?
 - “phone” → iPhone or Android?

Smart Behavior

- Detects topic shift and resets context
- Prevents conflicting intents
- Provides follow-up suggestions

Result

Chat behaves like a **real sales assistant**

3.4 AI Listing Generator (Seller Side)

Features

- Generate listing from:
 - images
 - short description
- Outputs:
 - title (EN + AR)
 - description
 - category
 - price range



Quality System

- Listing Quality Score (0–100)
- Suggestions:
 - improve title
 - add images
 - improve description

Result

- Removes friction for sellers
 - Increases listing quality across platform
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3.5 Search Intelligence & Ranking Optimization

Improvements

- Reduced semantic dominance
- Strong intent alignment:
 - cheap → penalize premium
 - luxury → penalize cheap
- Category correctness enforced
- Price normalization added

Outcome

- Top results are now **highly relevant**
 - Intent-based queries perform extremely well
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3.6 Behavioral Learning System

New Component

UserEvent tracking system



Tracks:

- searches
- clicks
- chat interactions

Ranking Enhancement

- Behavioral boost added:
 - based on clicks
 - logarithmic scaling (safe from abuse)

Result

System now:

Learns from real user behavior automatically

3.7 Admin Intelligence Dashboard

Features

- Top search queries
- Zero-result queries (inventory gaps)
- Top clicked products
- Chat intent insights

Key KPI Metrics

- Total searches
- Total clicks
- Zero-result rate
- Engagement rate



Result

Admin now has **full visibility of platform demand**

3.8 Admin Action Engine

Capabilities

Product Boosting

- Admin can manually boost listings
- Controlled with safety caps

Weak Listing Detection

- Identifies:
 - low completenessScore
 - missing images

Seller Intelligence

- Tracks:
 - top sellers
 - average quality scores

Result

Admin can:

Control marketplace supply, quality, and ranking in real time



3.9 Growth & Onboarding System

Buyer Onboarding

- “Try asking” quick prompts:
 - cheap car
 - iPhone
 - villa for rent

Seller Onboarding

- “AI Autofill” messaging
- Reduced friction for listing creation

Growth Hooks

- Shareable product links
- Invite seller tool (based on demand gaps)

Result

- Eliminates confusion
 - Improves activation rate
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3.10 Marketplace Safeguards

Empty State Solution

- No search returns empty
- Fallback items always shown

Seed Data System

- Script generates high-quality listings
- Ensures strong first impression



Result

- Prevents user drop-off
 - Ensures perceived platform quality
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4. Current System Capabilities

SellSell is now:

For Users

- Understands natural language
- Provides accurate results
- Supports conversational search

For Sellers

- Generates listings automatically
- Improves listing quality
- Reduces effort

For Admin

- Monitors real demand
 - Controls ranking
 - Identifies supply gaps
 - Boosts products instantly
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5. Key Achievements

- Built full AI-powered marketplace system
- Implemented hybrid semantic search
- Created real-time learning loop
- Designed admin control system
- Eliminated empty marketplace risk
- Achieved production-ready backend



6. Current Status



Production-Ready (Backend & Intelligence)

- Search system stable
- Chat system stable
- Ranking optimized
- Analytics operational
- Admin tools functional



Next Phase Required

- Real users
- Real sellers
- Real data flow

7. Remaining Gaps (Non-Technical)

- Marketplace liquidity (buyers + sellers)
- Real-world usage validation
- Seller onboarding at scale
- Growth and acquisition strategy

8. Final Assessment

System Maturity:

High (Production-ready backend)



Innovation Level:

Very High

- AI-native marketplace
- Conversational commerce
- Self-learning ranking system

Readiness:

Beta Launch Ready

9. Next Phase

Focus shifts from:

building features

to:

activating marketplace

Immediate priorities:

- onboard first sellers
 - onboard first users
 - monitor dashboard
 - refine based on real behavior
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Conclusion



SellSell has successfully evolved from a concept into a **fully functional AI-powered marketplace platform** with:

- intelligent search
- conversational interface
- automated listing generation
- self-learning ranking system
- admin control and analytics

The platform is now ready to enter **real-world beta testing and user acquisition phase**.